

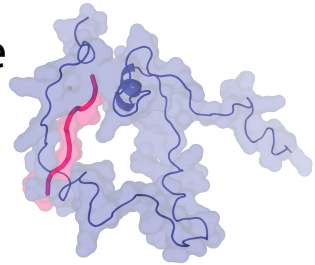
CIS 6270

Module 2: Flow Matching

Lecture 2.2: Flow Matching
Thu, Sep 10, 2026

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Integrating from time 0 to time t

$$\log p_t(x_t) = \log p_0(x_0) - \int_0^t \nabla \cdot v_s(x_s)\, ds$$

The log density log $p_t(x_t)$ at time t equals the initial log density log $p_0(x_0)$ minus the velocity divergence $\nabla \cdot v_s(x_s)$ accumulated along the trajectory from time 0 to time t.

This equation is the continuous-time counterpart of the change-of-variables formula from Lecture 1.

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The determinant records the accumulated volume change over a finite time interval. The divergence records the instantaneous rate of volume change.

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The continuity-equation derivation and the Jacobian-determinant derivation therefore produce the same log-density equation.

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The final density $p_1(x_1)$ equals one-half of the initial density $p_0(x_0)$ because the flow map doubles local lengths.

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The standard deviation doubles, local lengths double, and the density at corresponding points decreases by one-half.

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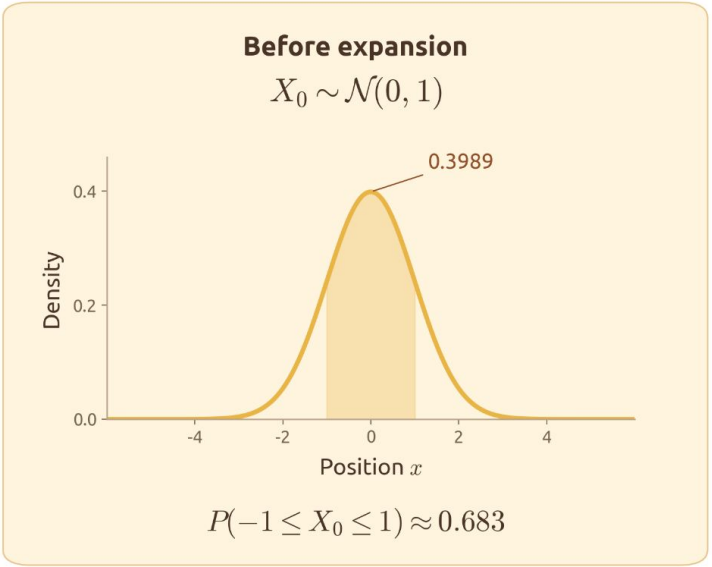
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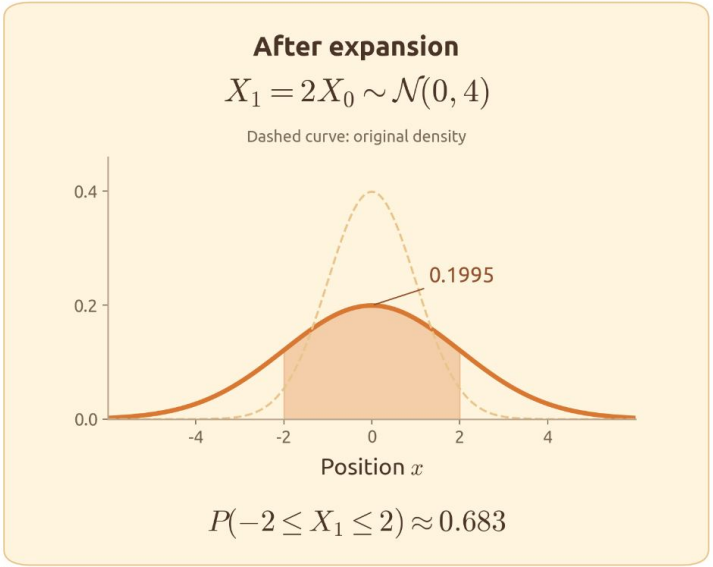
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Doubling every position spreads the distribution over twice the width and halves its density, while preserving the probability within corresponding intervals.





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The rate of change of the total probability mass obtained by integrating $p_t(x)$ over the complete state space equals zero.

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The continuity equation redistributes probability through the state space while preserving the total amount of probability.

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The neural network determines the velocity.

The ODE solver constructs the trajectory.

The divergence determines the accompanying density change.

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The flow map and its Jacobian describe the same volume change over a finite interval.

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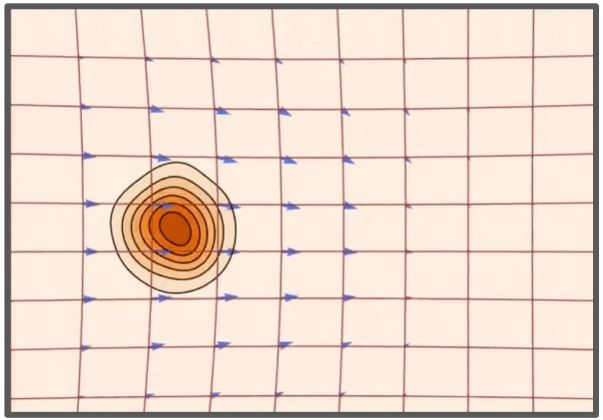
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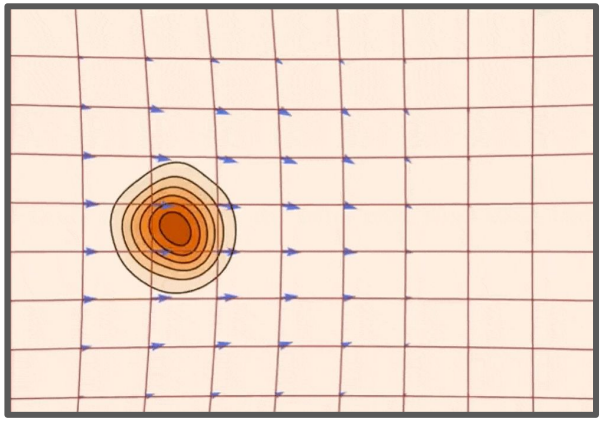
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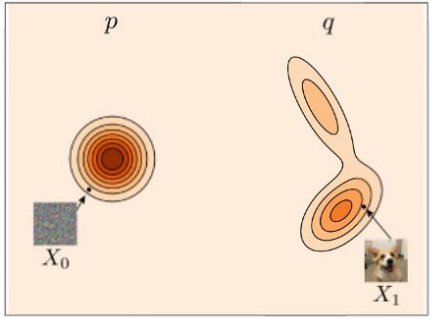
We need a training strategy that begins with a chosen path between prior samples and data samples, computes the velocities along that path, and learns those velocities through regression

Flow Matching

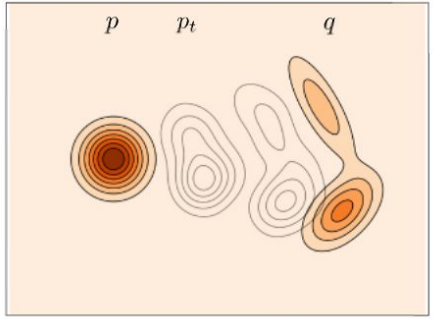


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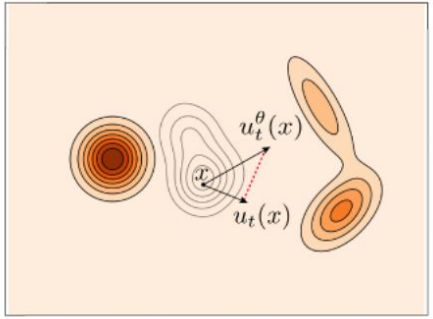




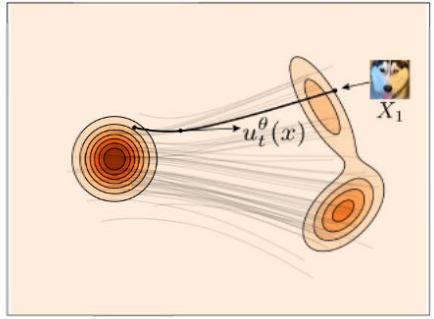
(a) Data.



(b) Path design.



(c) Training.



(d) Sampling.

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We now need to pair prior samples with data samples.

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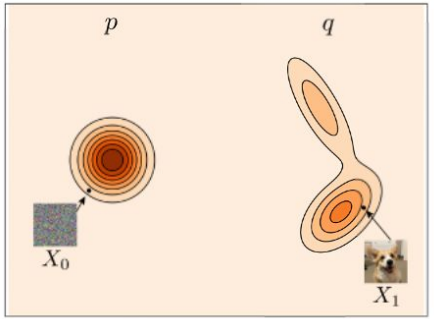
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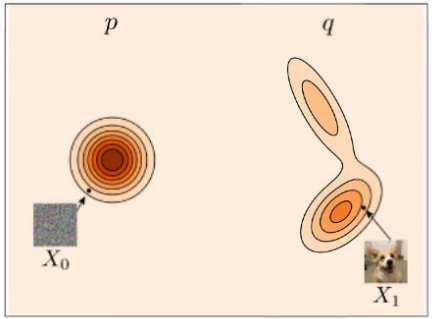
Coupling the endpoints



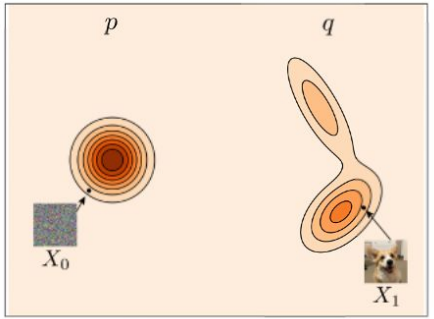
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$$(X_0, X_1) \sim \pi$$



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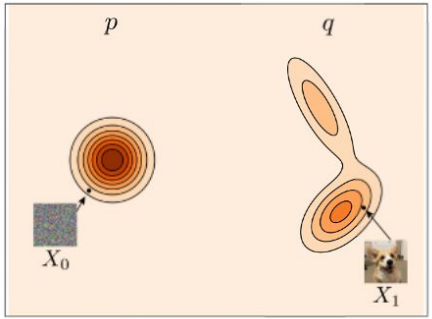
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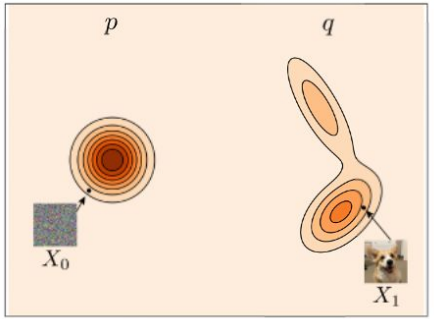
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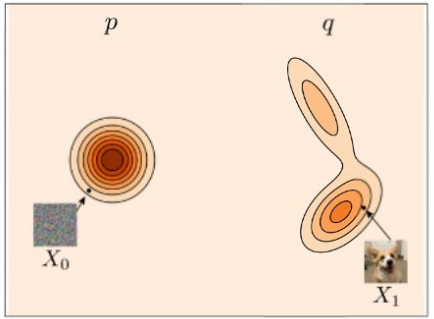
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Under this coupling, we draw a Gaussian point and a data point independently and connect them.

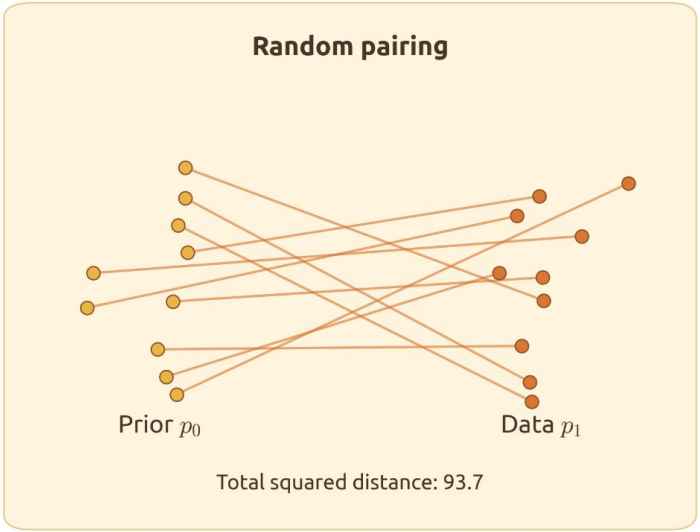
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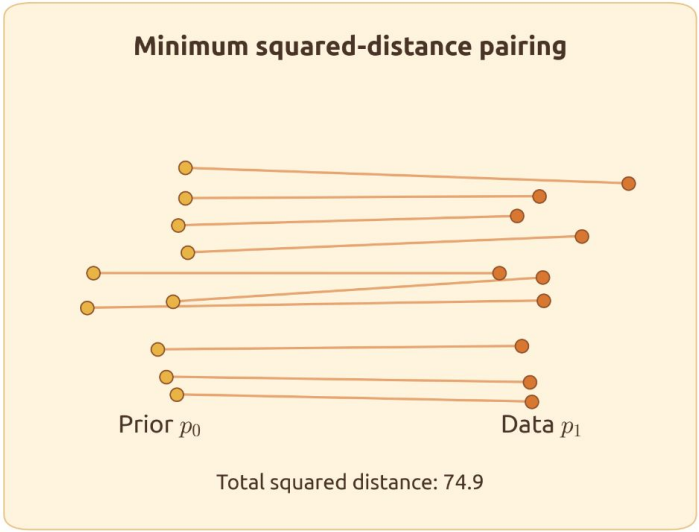
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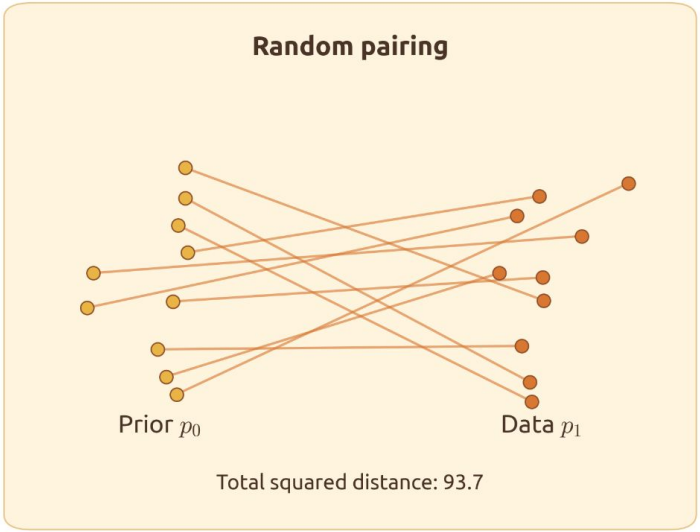
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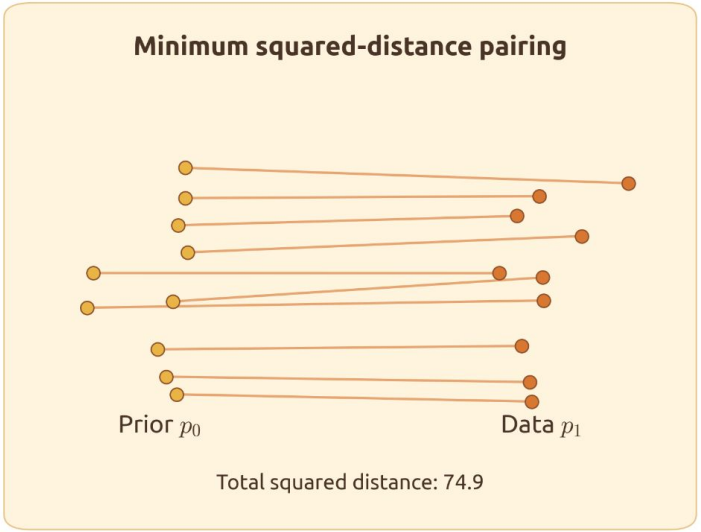




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Once we have an endpoint pair, we need a continuous path connecting it.

Interpolation between prior and data

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The simplest path is a straight line

$$\boxed{X_t = (1-t)X_0 + tX_1, \qquad 0 \le t \le 1}$$

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The explicit interpolation tells us every point along the path

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We can therefore differentiate it to obtain the required velocity.

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Its value remains constant along a linear interpolation because a straight path covers equal distances during equal time intervals.

Numerical example with one endpoint pair

Choose the one-dimensional endpoints

$$x_0 = -2, \qquad x_1 = 3$$

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At several times, you can see what x_t is

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Training includes many independently sampled paths, and their intermediate states can overlap!

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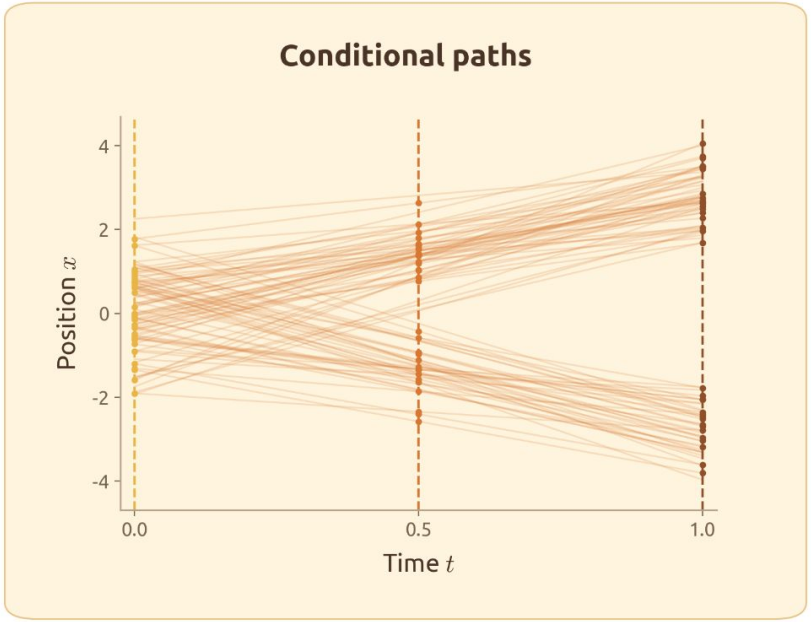
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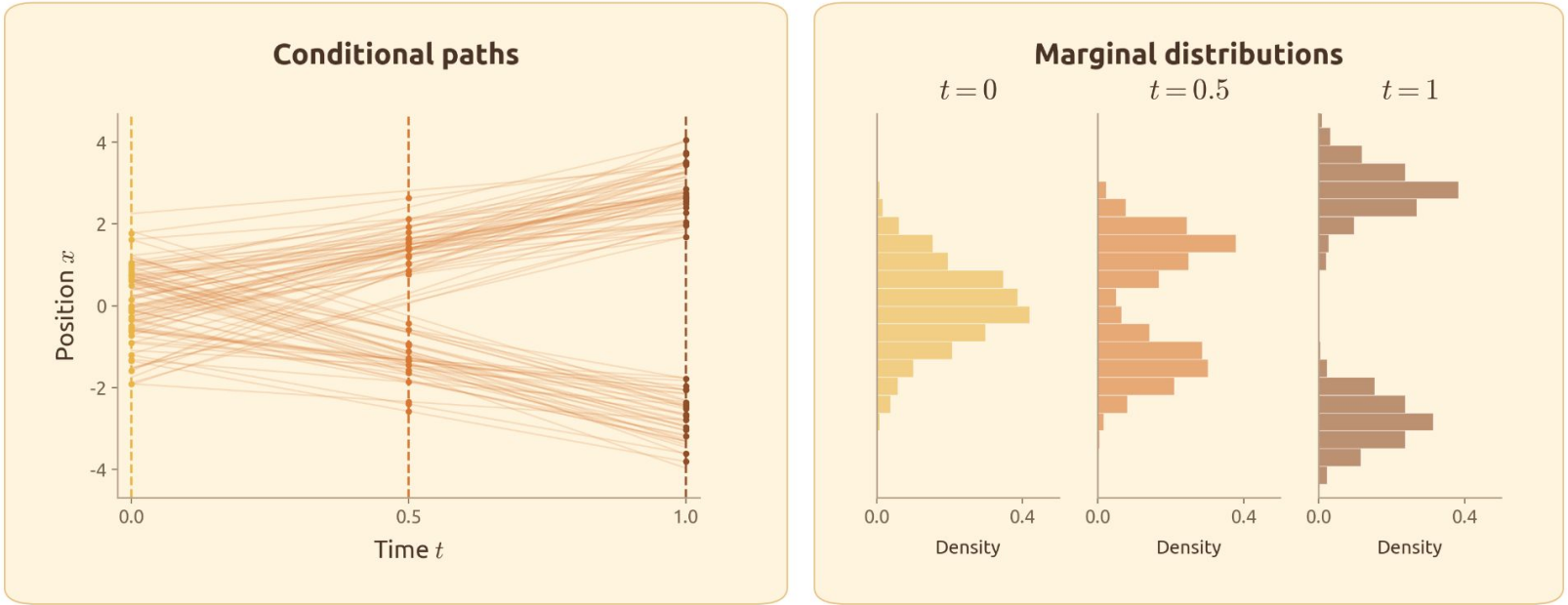
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Each line follows one noise-data pair



Each line follows one noise-data pair; collecting all positions at a particular time produces the distribution shown in that time's histogram.



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The conditional construction gives one target for each sampled path. The marginal field combines those targets into one deterministic velocity at every state and time.

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Sampling after training follows this marginal field. Generated trajectories do not need to reproduce the individual straight training paths. Their distribution must reproduce the marginal density path.

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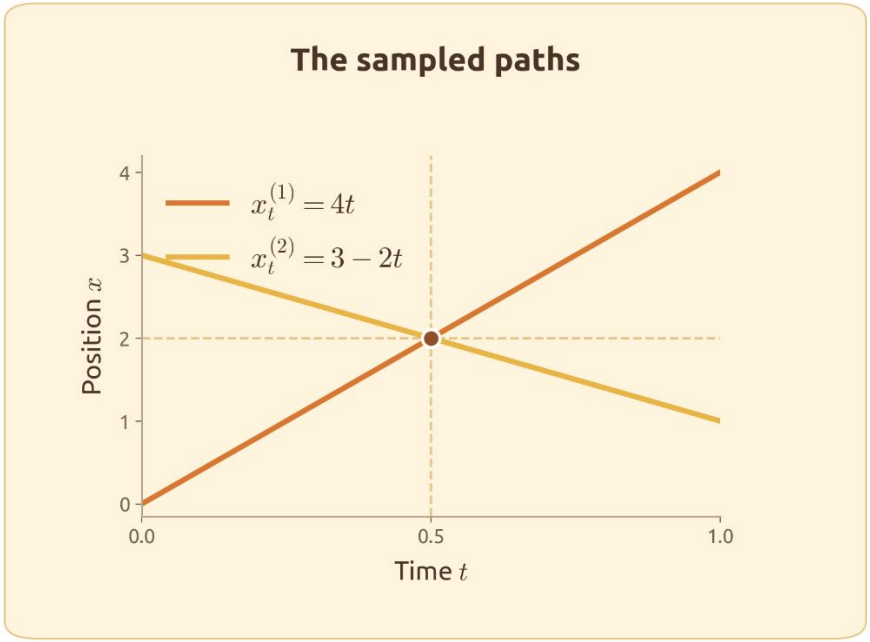
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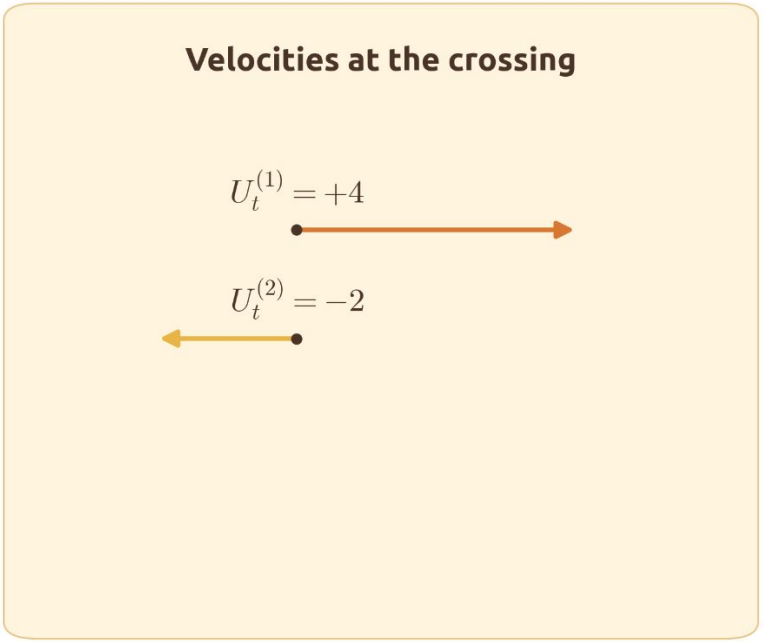
The marginal velocity equals 1 because the two equally likely conditional velocities 4 and -2 average to 1.

A deterministic velocity field can assign only one velocity to x = 2 at t = 0.5. The average velocity reproduces the correct net movement of probability through that location.

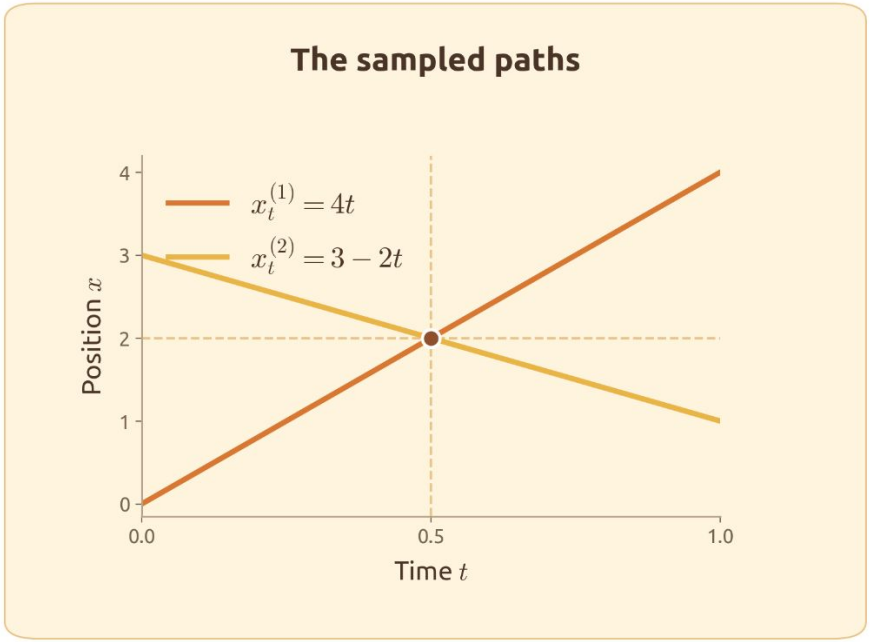
This conditional average also appears automatically when we minimize squared prediction error.

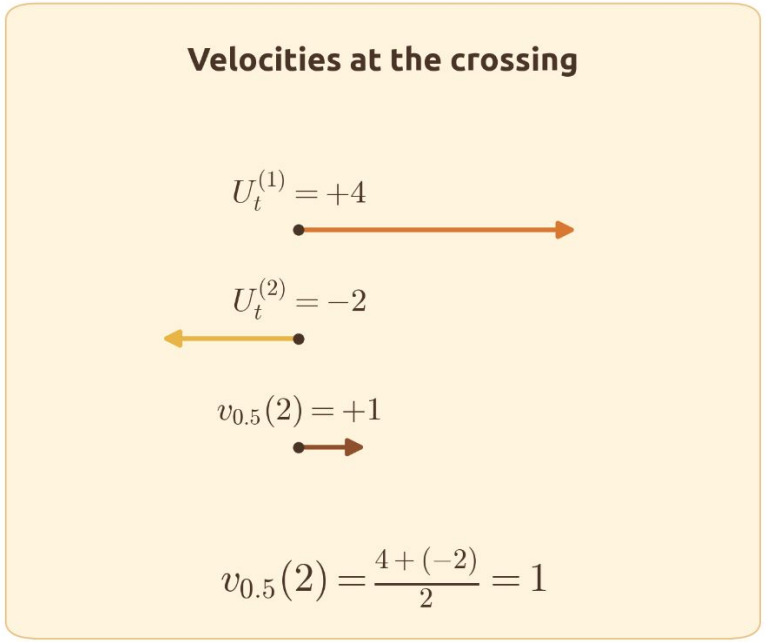
Two paths reach the same position at the same time with velocities +4 and –2



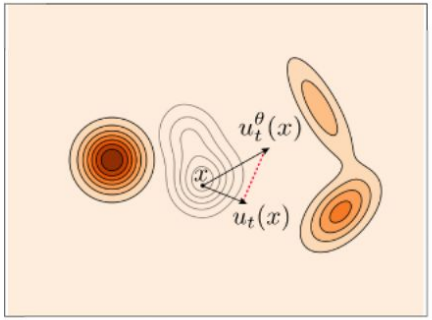


Two paths reach the same position at the same time with velocities +4 and –2; because they are equally likely, the best single velocity prediction is their average, +1.

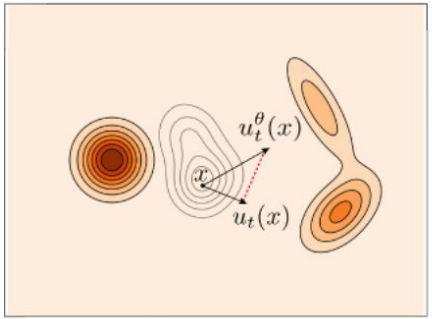




The flow-matching regression objective



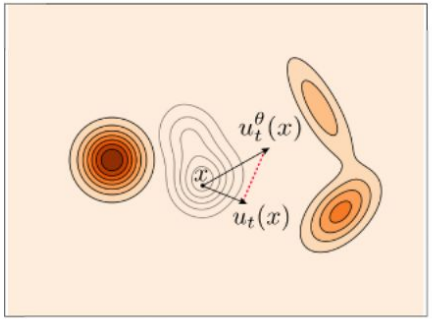
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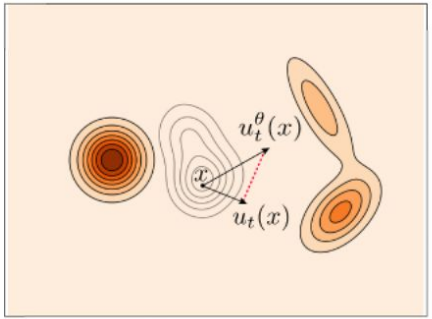
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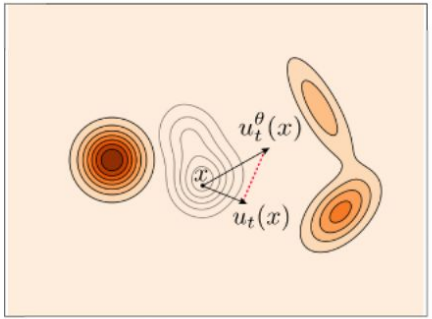
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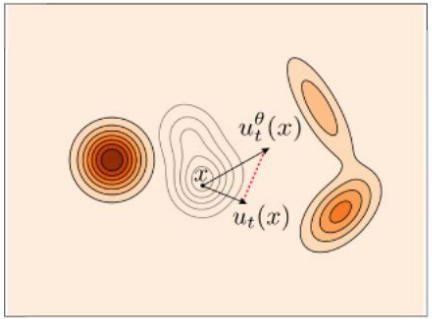
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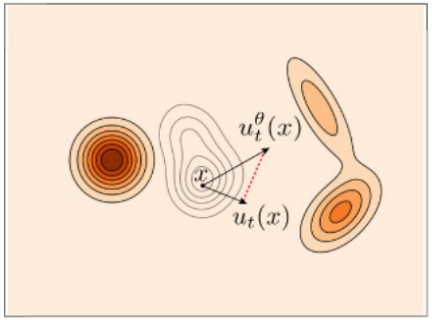
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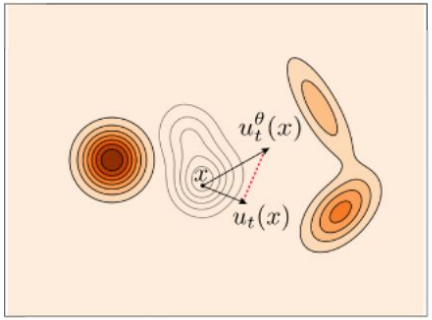
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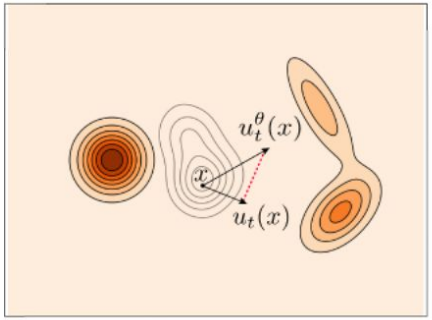
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The value agrees with the conditional average:

$$\frac{4 + (-2)}{2} = 1$$

Conditional Flow-Matching Equivalence Theorem

The ideal marginal flow-matching objective would compare the network with the marginal velocity $v_t(X_t)$:

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Training against easily computed conditional velocities produces the same parameter updates as training against the generally unavailable marginal velocity field.

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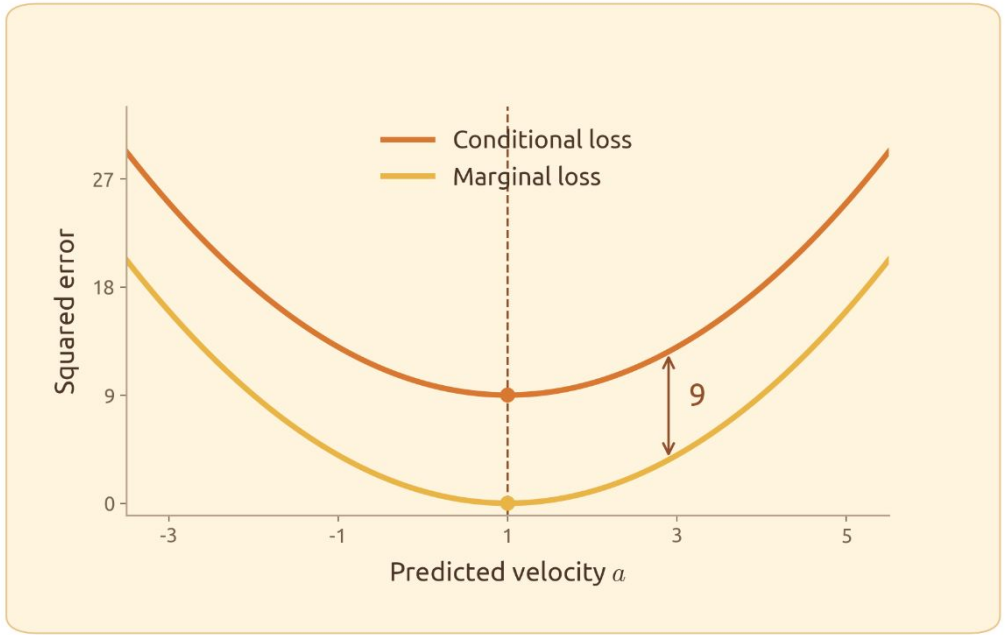
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The average squared error against the two individual targets equals the squared error against their mean (+ 9).

The conditional loss sits 9 units higher, but both have identical slopes and are minimized at a = 1. The value 9 is the target variance in this particular example.



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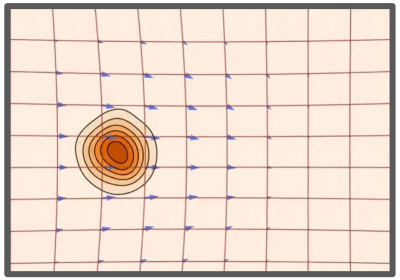
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So the terminal distribution p_1^θ generated by the learned velocity field equals the data distribution p_{data} when the learned velocity matches the target marginal velocity exactly.

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Therefore

$$X_t \sim \mathcal{N}(4t, 1)$$

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The particle trajectories, marginal densities, conditional velocities, and learned ODE all agree.

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This form will also help connect flow matching to diffusion, where signal and noise contributions follow time-dependent schedules.

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The sample follows the same straight segment, beginning slowly and accelerating toward the data.

Halfway through the time interval, it has covered one-quarter of the distance.

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The direction stays fixed, while the speed increases linearly with time. We train the network against this time-dependent target.

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For independent, centered endpoints with identity covariance, this schedule preserves identity covariance throughout the interpolation

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For either alternative, construct the intermediate state, differentiate its interpolant, and use that derivative as the target in the same squared-error training procedure.

A quick numerical comparison:

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At t = 0.25, linear interpolation has covered 25% of the straight segment, quadratic interpolation has covered only 6.25% because progress follows t_2, and the sine/cosine weights produce a curved path with a different velocity direction.

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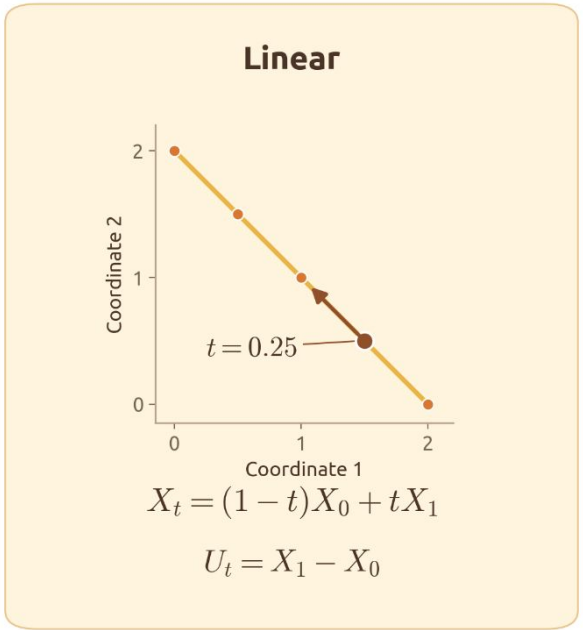
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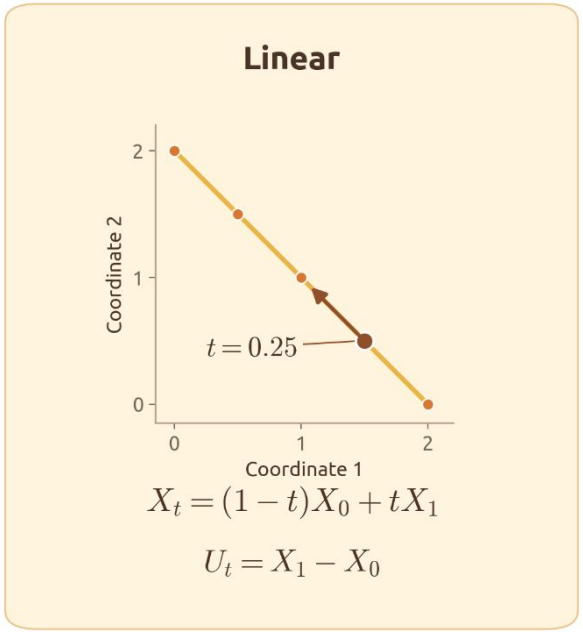
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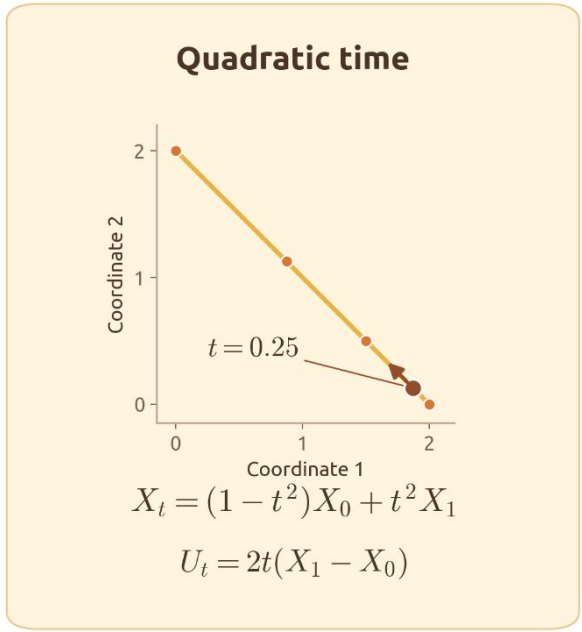
Trying different interpolants can help when velocities are difficult to learn or sampling requires many integration steps, because each choice changes the intermediate distributions and velocity targets.

Linear: The point moves along a straight line at constant speed, covering equal distances in equal time intervals



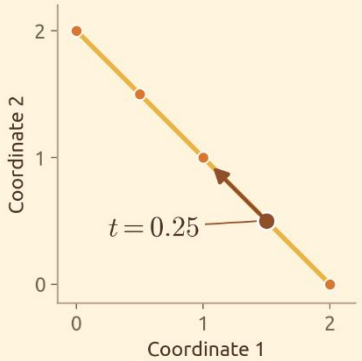
Quadratic time: The point follows the same straight line but accelerates, so successive time markers become farther apart.





Sine/cosine: The point follows a curved path, with its velocity continually changing direction.

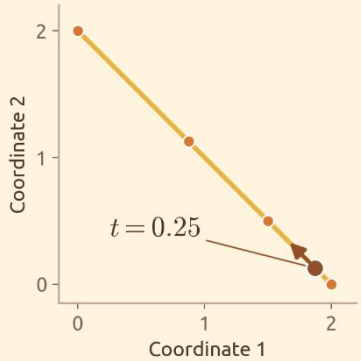
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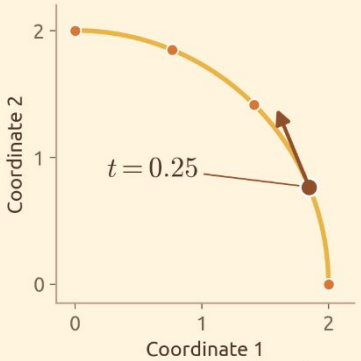
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$$X_t = \cos(\tfrac{\pi t}{2})X_0 + \sin(\tfrac{\pi t}{2})X_1$$

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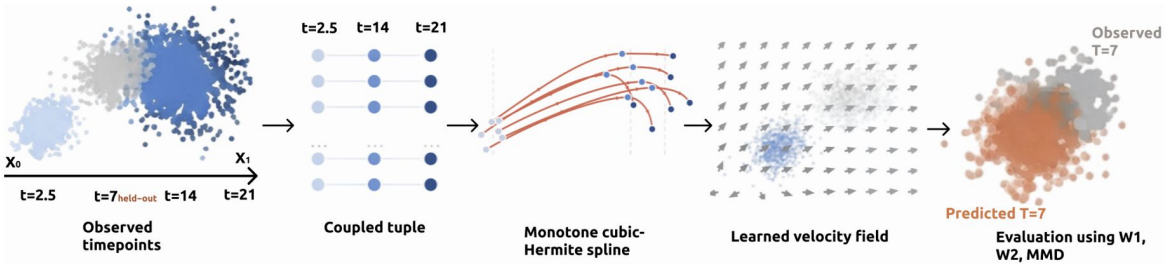
We have been doing this for unique data modalities

CHIME: Cubic-Hermite Interpolation of Marginal Evolution from Sparse T-Cell Snapshots

Aastha Pal, Joseph A. Fraietta, Pranam Chatterjee

29 Aug 2026 (modified: 02 Sept 2026) NeurIPS 2026 Workshop AI4Science Submission
AI4Science, Area Chairs, Reviewers, Authors Revisions CC BY 4.0

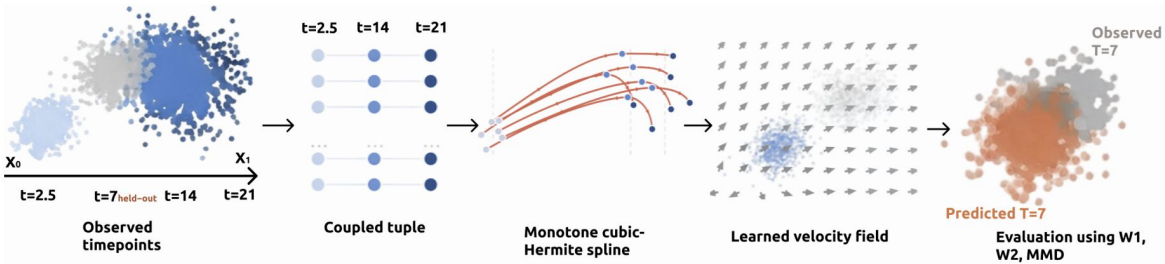
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For one interval rescaled to t ∈ [0, 1], CHIME's cubic-Hermite interpolant is:

$$X_t = (2t^3 - 3t^2 + 1)X_0 + (-2t^3 + 3t^2)X_1 + (t^3 - 2t^2 + t)M_0 + (t^3 - t^2)M_1,$$

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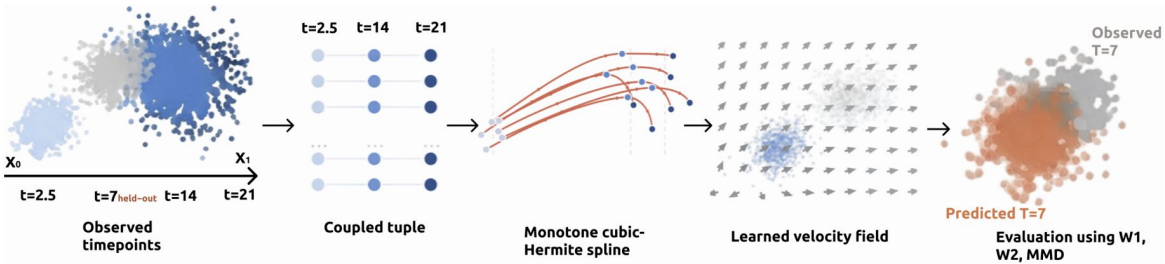


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where M_0 and M_1 are the endpoint slopes multiplied by the original interval duration.

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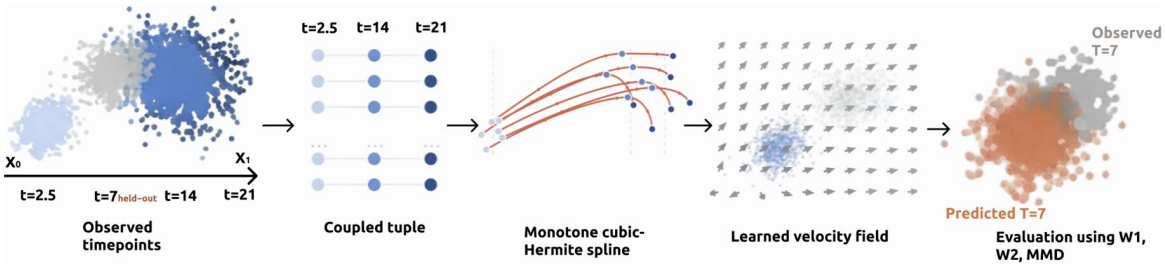


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The intermediate state X_t combines endpoints X_0 and X_1 and slope terms M_0 and M_1 through cubic time weights, with slopes chosen to connect observed cell states smoothly and limit overshoot.

Worked better than base methods!

Table 1: Held-out day-7 population prediction in the initial chronic-LCMV study. Day 7 was excluded from all fitting, and predictions were initialized from the observed day 2.5 population. Values are the mean $\pm$ standard deviation across five random seeds; lower is better.

Model	$W_1 \downarrow$	$W_2 \downarrow$	MMD $\downarrow$
No-move	17.67 ± 0.09	18.14 ± 0.09	0.625 ± 0.008
OT-CFM (linear)	12.28 ± 0.07	12.56 ± 0.07	0.328 ± 0.007
MMFM	11.18 ± 0.15	11.40 ± 0.15	0.204 ± 0.008
PRESCIENT	18.22 ± 3.49	18.71 ± 3.38	0.466 ± 0.125
moscot	12.15 ± 0.12	12.41 ± 0.11	0.315 ± 0.009
CHIME (ours)	$\mathbf{10.60 \pm 0.15}$	$\mathbf{10.78 \pm 0.16}$	$\mathbf{0.169 \pm 0.009}$

CHIME achieved the lowest mean error across all three metrics (Table 1). It reduced W_2 by 14.2% relative to linear OT-CFM and by 5.4% relative to MMFM, the strongest competing method on this metric. Relative to the no-move control, CHIME reduced W_2 by 40.6%. The small variation across seeds indicated that the day-7 reconstruction was stable across training runs.

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Fifth, update θ using

$$\left\| v_\theta(X_t, t) - U_t \right\|^2$$

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During training, we can sample any time t and construct X_t directly. This avoids simulating the ODE to create each training example.

How do you generate with a trained flow?